Dynamically decoupled radio-frequency(DDrf) Gate

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Quantum Info with Materials and Engineering

Outline

1. **DDrf Gate**
    - Hamiltonian engineering & conditional rotation
    - DDrf spectroscopy
    - Hybrid DDrf gate
2. **Side-Peak Problem**
    - Observation in numerics
    - Detuned-Rabi explanation
3. **Suppression**
    - Apodized pulses & window comparison
    - Outlook: time-dependent Ω_{RF} via Magnus expansion

DDrf: Hamiltonian Engineering

DDrf = selective, phase-controlled RF driving of nuclear spins, interleaved with dynamical decoupling on the electron spin.

If the Hamiltonian is **block-diagonal** in the electron basis,

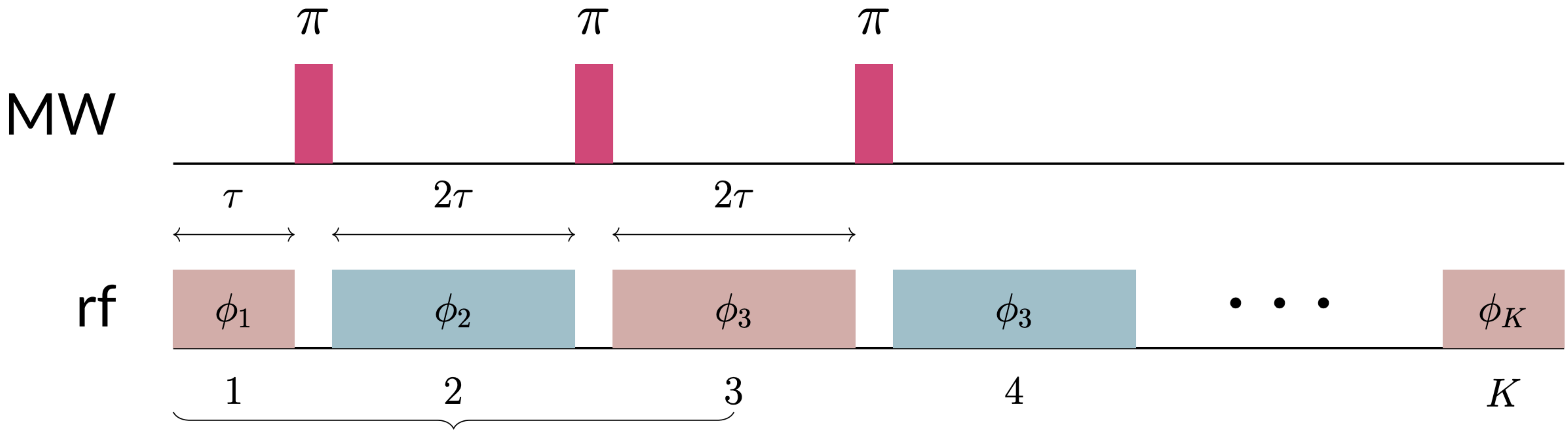
$$H = |0\rangle\langle 0| \otimes H_0 + |1\rangle\langle 1| \otimes H_1,$$

then sandwiching evolution with an electron π-pulse swaps the two branches:

$$|+\rangle|0\rangle \xrightarrow{t_1,\,\pi,\,t_2} \frac{1}{\sqrt{2}}\Big(|0\rangle \otimes \underbrace{e^{-iH_0t_2}e^{-iH_1t_1}}_{U_0}|0\rangle + |1\rangle \otimes \underbrace{e^{-iH_1t_2}e^{-iH_0t_1}}_{U_1}|0\rangle\Big).$$

Generally $U_0 \neq U_1$ — a **conditional gate**.

DDrf: Pulse Sequence



$$(\tau\text{-}\pi\text{-}2\tau\text{-}\pi\text{-}\tau)^{N/2}$$

odd k: resonant for $|1\rangle$ $(m_s = -1)$

even k: resonant for $|0\rangle$ $(m_s = 0)$

DDrf Spectroscopy: NV–^{13}C Hamiltonian

For NV–^{13}C with rf driving:

$$\begin{aligned} H &= |0\rangle\langle 0| \otimes H_0 + |-1\rangle\langle -1| \otimes H_1 \\ H_0 &= \omega_0 I_z + 2\Omega_{\mathrm{RF}} \cos(\omega_{\mathrm{RF}} t + \phi)\, I_x \\ H_1 &= \omega_1 \tilde{I}_z + 2\Omega_{\mathrm{RF}} \cos\beta \cos(\omega_{\mathrm{RF}} t + \phi)\, \tilde{I}_x + 2\Omega_{\mathrm{RF}} \sin\beta \cos(\omega_{\mathrm{RF}} t + \phi)\, \tilde{I}_z \end{aligned}$$

The $|1\rangle$ branch has a **tilted** nuclear quantization axis (angle β, with $\sin\beta = A_\perp/\omega_1$). This small tilt is what enables hyperfine-mediated control of nuclei with vanishing $A_\perp$.

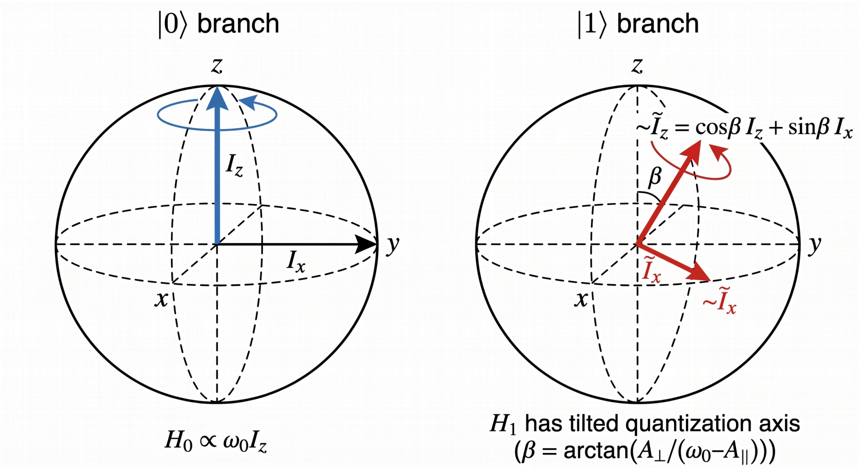
DDrf Spectroscopy: Rotating Frame

Two electron-conditioned rotating frames, $R_s(t) = e^{i\omega_{\mathrm{RF}} t\, \tilde{I}_z^{(s)}}$, give:

$$H_0' = (\omega_0 - \omega_{\mathrm{RF}})\, I_z + \Omega_{\mathrm{RF}}(\cos\phi\, I_x + \sin\phi\, I_y)$$

$$H_1' = (\omega_1 - \omega_{\mathrm{RF}})\, \tilde{I}_z + \Omega_{\mathrm{RF}} \cos\beta\, (\cos\phi\, \tilde{I}_x + \sin\phi\, \tilde{I}_y)$$

At resonance ($\omega_{\mathrm{RF}} = \omega_1$) and approximation to $\omega_0 - \omega_{\mathrm{RF}} \gg \Omega_{\mathrm{RF}}$ and $\beta \to 0$, H_1' is a pure transverse drive, while H_0' is pure Z-rotation.



DDrf Spectroscopy: Full Time Evolution

Each MW π-pulse acts as an **instantaneous frame swap** $\Lambda_{s,\bar{s}}(t) \equiv R_s(t)R_{\bar{s}}(t)^\dagger$ between the two rotating frames. The full unitary $U = \sum_s |s\rangle\langle s| \otimes U_s$ is then a chain of H_s'-segments separated by frame swaps:

$$U_s = R_s(4N\tau)^\dagger \cdot e^{-iH_s'\tau} \cdot \Lambda_{s,\bar{s}}((2N-1)\tau) \cdot e^{-iH_{\bar{s}}'2\tau} \cdot \Lambda_{\bar{s},s}((2N-3)\tau) \cdot e^{-iH_s'\tau} \cdots$$
$$\cdots e^{-iH_s'\tau} \cdot \Lambda_{s,\bar{s}}(3\tau) \cdot e^{-iH_{\bar{s}}'2\tau} \cdot \Lambda_{\bar{s},s}(\tau) \cdot e^{-iH_s'\tau} \cdot R_s(0).$$

Exact under the assumption of negligible MW pulse duration, and **much faster than direct Schrödinger integration** when Ω_{RF} is constant — each $e^{-iH_s'\tau}$ is a single matrix exponential of a time-independent Hamiltonian.

DDrf Spectroscopy: Procedure & Results

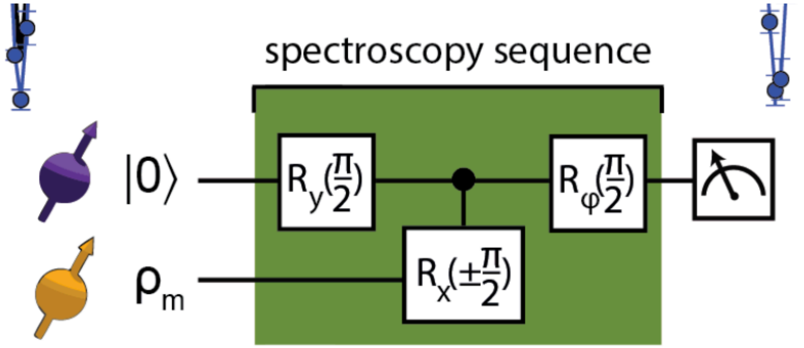
Sequence: $\pi/2 \to \mathrm{DDrf}(N,\tau) \to \pi/2_\phi$, projecting onto $|+\rangle$ ($\phi = \pi/2$).

For N nuclear spins:

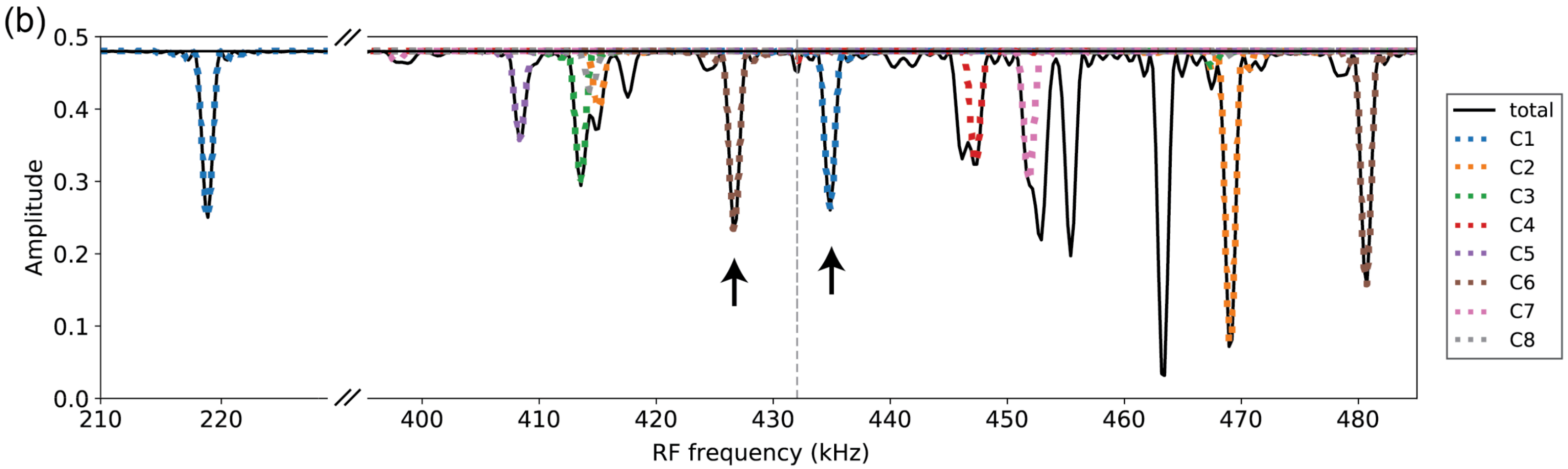
$$P_x = \frac{1}{2} + \frac{1}{2^{N+1}} \Re \operatorname{Tr} U_0 U_1^\dagger,$$

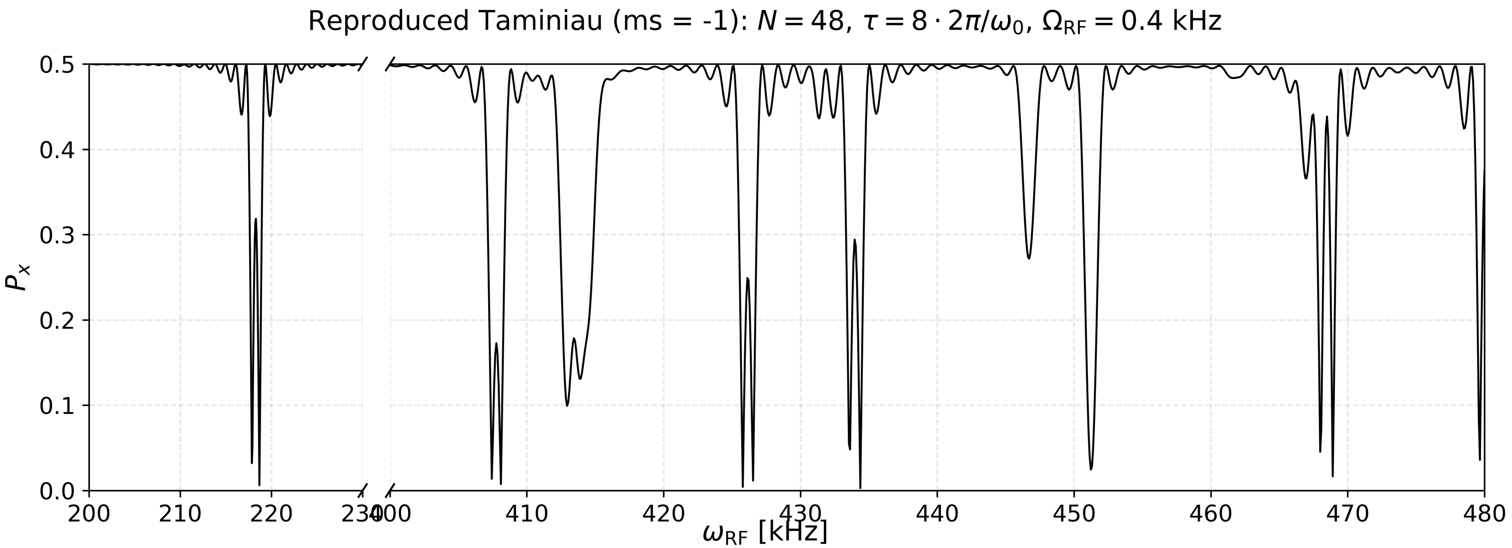
$$\operatorname{Tr} U_0 U_1^\dagger = \prod_i \operatorname{Tr} U_0^i U_1^{i\,\dagger}.$$

Peaks appear at $\omega_{\mathrm{RF}} = \omega_1$.



DDrf Spectroscopy: Reproduced Taminiau Results

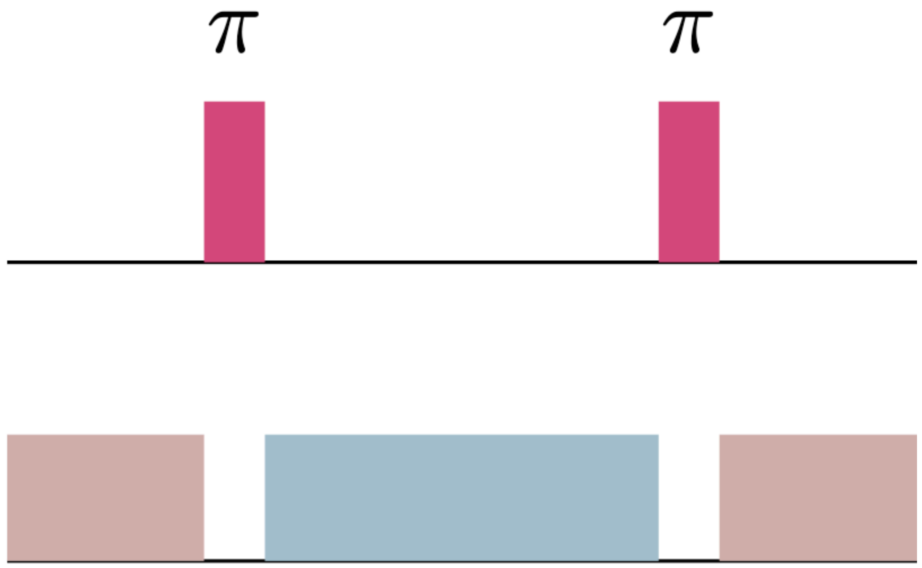




DDrf: Per-Cell Decomposition

The DDrf sequence is built from identical 4-τ cells. Each cell is itself block-diagonal:

$$V^{(k)} = |0\rangle\langle 0| \otimes V_0^{(k)} + |1\rangle\langle 1| \otimes V_1^{(k)}, \qquad U = \sum_{s\in\{0,1\}} |s\rangle\langle s| \otimes \prod_{k=1}^{N/2} V_s^{(k)}.$$



DDrf: Commutation Trick

A z-rotation can be commuted through a transverse rotation by shifting its azimuthal angle:

$$e^{i\alpha I_z}\,(\cos\phi\, I_x + \sin\phi\, I_y)\, e^{-i\alpha I_z} = \cos(\phi-\alpha)\, I_x + \sin(\phi-\alpha)\, I_y.$$

Applied per cell (Taminiau limit, $\beta = 0$, $\omega_{\mathrm{RF}} = \omega_1$):

$$V_0^{(k)} = e^{-iH_0'\tau}\, e^{-iH_1' 2\tau}\, e^{-iH_0'\tau} \;=\; e^{-i\,2\delta_0\tau\, I_z}\, e^{-i\,2\Omega\tau\,\hat{\phi}_k'\cdot\vec{I}}.$$

Each cell splits cleanly into a z-piece + a transverse rotation. Choosing ϕ_k to align successive transverse axes makes the product over k contracted into a single conditional rotation.

Hybrid DDrf: Two Limits and Their Combination

The per-cell picture exposes two distinct mechanisms that both produce conditional rotation:

- **CPMG limit** ($\Omega_{\mathrm{RF}} \to 0$): pure dynamical decoupling. At $\tau \simeq \frac{(2k-1)\pi}{2\omega_0 + A_\parallel}$, the change-of-frame between R_0 and R_1 alone yields a hyperfine-driven conditional rotation.
- **Taminiau limit** ($\beta \to 0$, $\omega_{\mathrm{RF}} = \omega_1$): pure RF driving. Conditional rotation comes from Ω_{RF} alone, independent of $A_\parallel$.

Hybrid idea. Choose τ at the CPMG resonance and add RF driving on top. The two mechanisms add coherently:

$$\theta_{\mathrm{cond}} \approx \underbrace{N\,\theta_{\mathrm{CPMG}}(\tau)}_{\text{frame-swap}} + \underbrace{N\,\Omega_{\mathrm{RF}}\tau}_{\text{RF drive}}$$

Hybrid DDrf: Taylor Expansion in β

The hybrid regime sits between the two clean limits, but β is small. Treat it as a perturbation around the Taminiau limit:

$$V_s^{(k)} \;\simeq\; V_s^{(k)}\Big|_{\beta=0} \;+\; \beta\,\frac{\partial}{\partial\beta}V_s^{(k)}\Big|_{\beta=0} \;+\; \mathcal{O}(\beta^2).$$

Limit checks (the reason this approach is trustworthy):

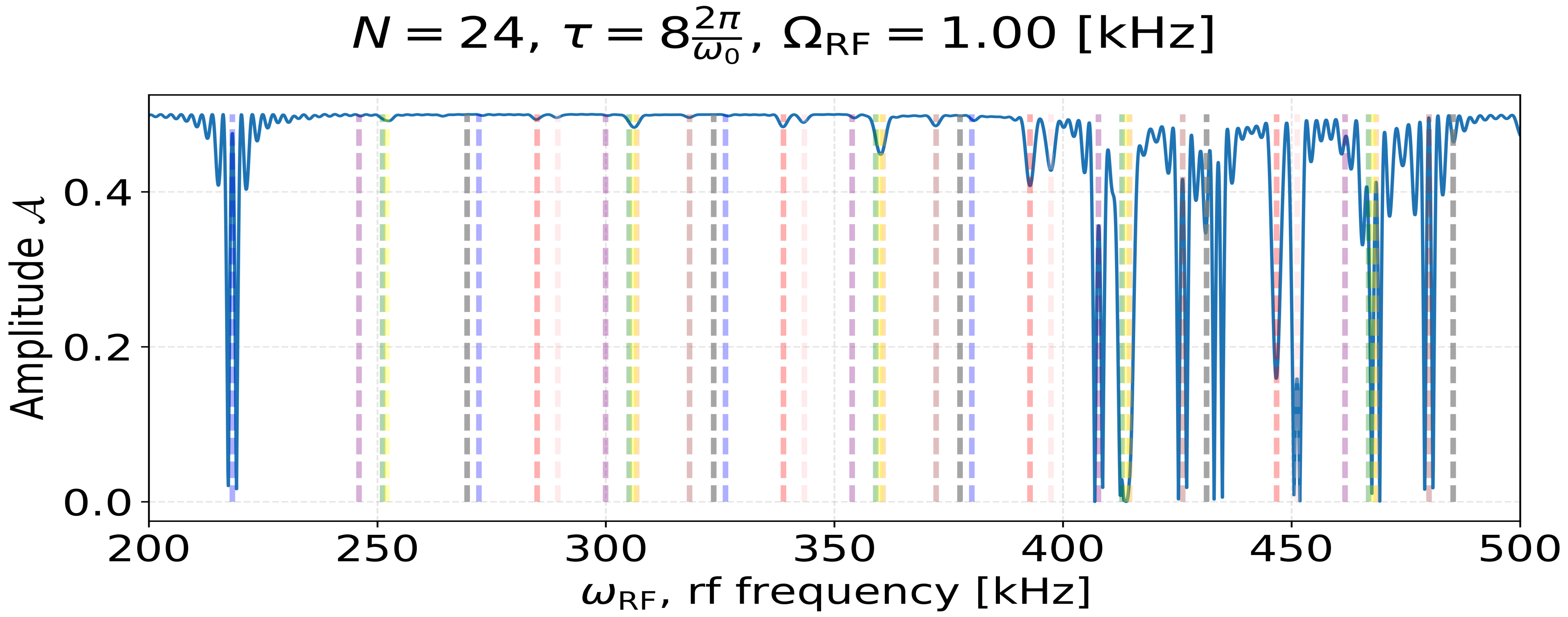
- $\beta \to 0$ — recovers the Taminiau result.
- $\Omega \to 0$ — recovers the CPMG mechanism.

Outcome (qualitative). The leading β-correction adds a small k-dependent rotation around an axis that mixes I_z, I_x, I_y. With an appropriate choice of ϕ_k and τ, these corrections build up — the effective rotation angle exceeds bare $\Omega\tau$, realizing the speed-up promised on the previous slide.

Side-Peak Problem: Observation

In the Taminiau limit, we have $U_s = R_z(N(\omega_L - \omega_1)\tau) \cdot R_\phi(\pm N\Omega_{\mathrm{RF}}\tau)$.

[Observation] When $N\Omega_{\mathrm{RF}}\tau = 2\pi$, $U_0 = U_1$ — the gate becomes **unconditional**, so a flat spectroscopy signal is expected. However, …



Side-Peak Problem: Detuned Rotating Frame

Restore finite detuning $\delta_1 = \omega_1 - \omega_{\mathrm{RF}}$ (assume $\beta = 0$ for clarity):

$$H_1' = \delta_1 I_z + \Omega(\cos\phi I_x + \sin\phi I_y) = \Omega_{\mathrm{eff}}\,\hat{n}(\phi)\cdot\vec{I},$$

$$\Omega_{\mathrm{eff}} = \sqrt{\Omega^2 + \delta_1^2}, \qquad \sin\gamma = \frac{\delta_1}{\Omega_{\mathrm{eff}}}, \qquad \hat{n}(\phi) = (\cos\gamma\cos\phi,\ \cos\gamma\sin\phi,\ \sin\gamma).$$

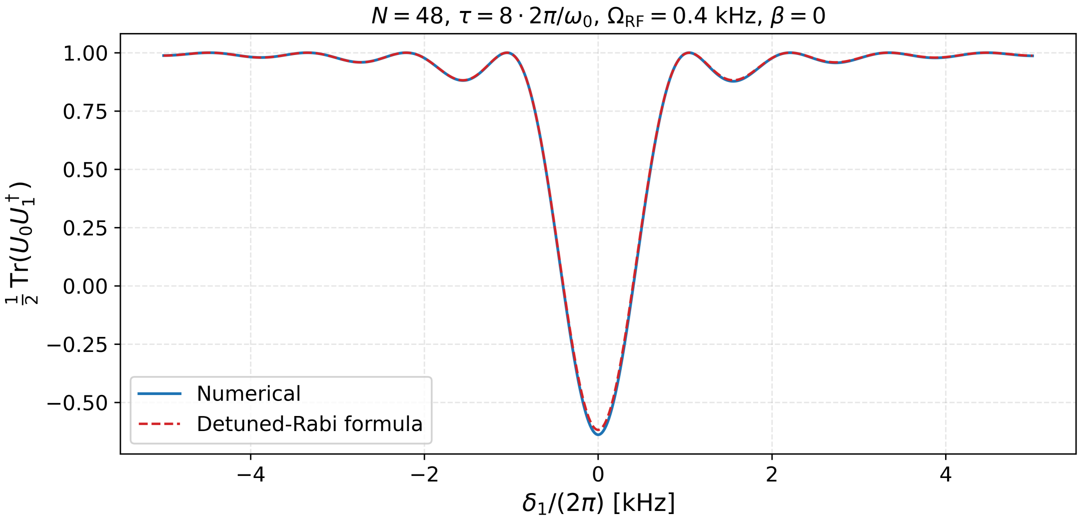
Conjugation still works because the tilt angle γ is invariant under z-rotations. Result:

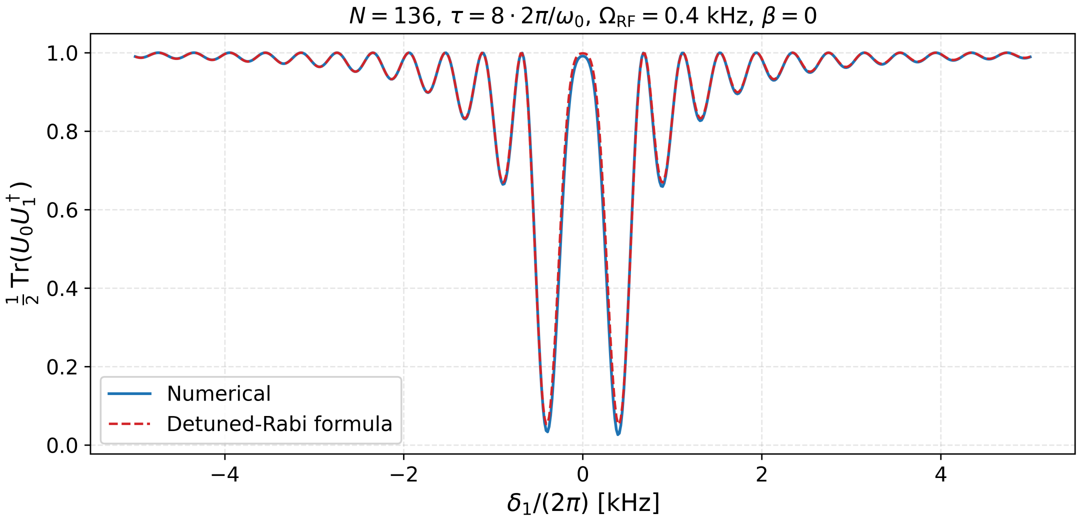
$$V_s^{\mathrm{tot}} = e^{-iN\delta_0\tau I_z}\,e^{-i\Omega_{\mathrm{eff}}N\tau\,\hat{n}_s\cdot\vec{I}}, \qquad \hat{n}_{0,1} = (\pm\cos\gamma,\ 0,\ \sin\gamma).$$

Side-Peak Problem: Detuned Rabi Formula vs. Numerics

$$\frac{1}{2}\,\mathrm{Tr}(U_0 U_1^\dagger) = 1 - \frac{2\Omega^2}{\Omega^2 + \delta_1^2}\,\sin^2\!\left(\frac{\sqrt{\Omega^2 + \delta_1^2}\,N\tau}{2}\right)$$

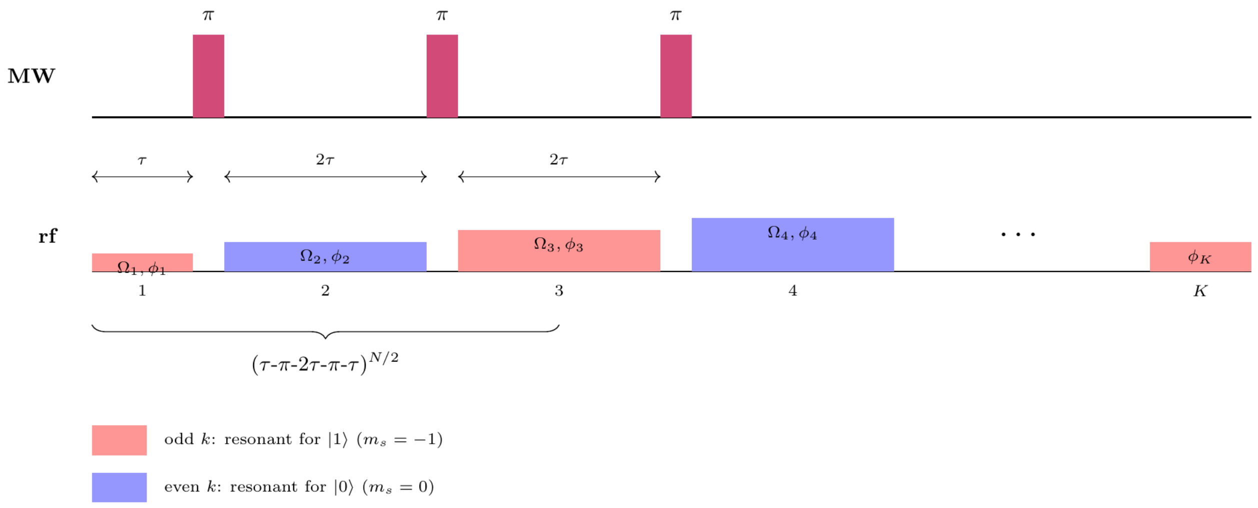
The detuned-Rabi formula reproduces both the envelope and the side-lobe period:

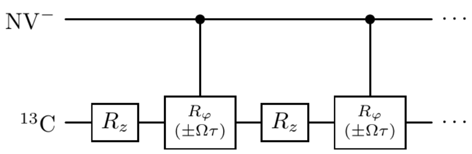




Suppression: Apodized Pulse Idea

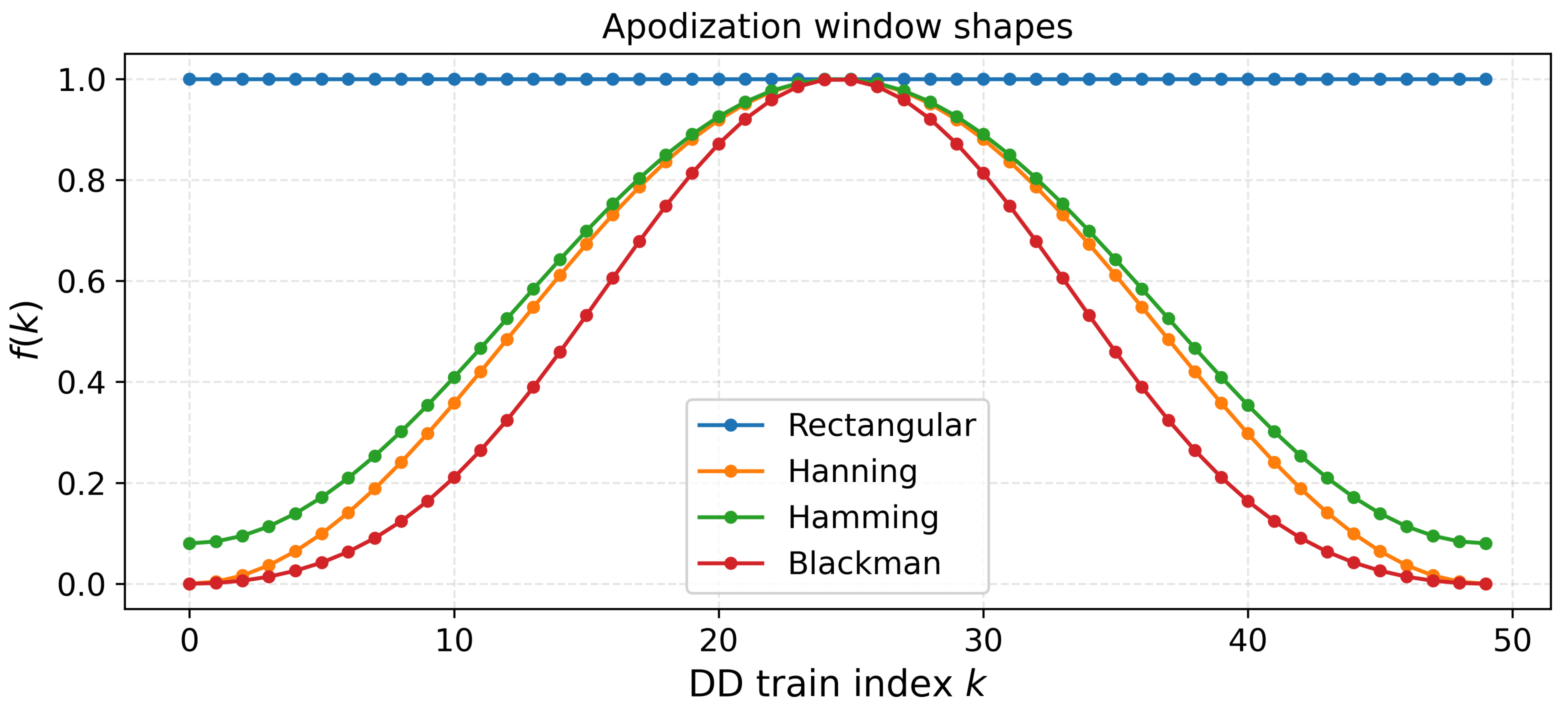
Replace constant RF amplitude with a per-cell envelope $\Omega_k = \Omega\, f(k)$, a discrete window function (Hanning, Hamming, Blackman, ...):





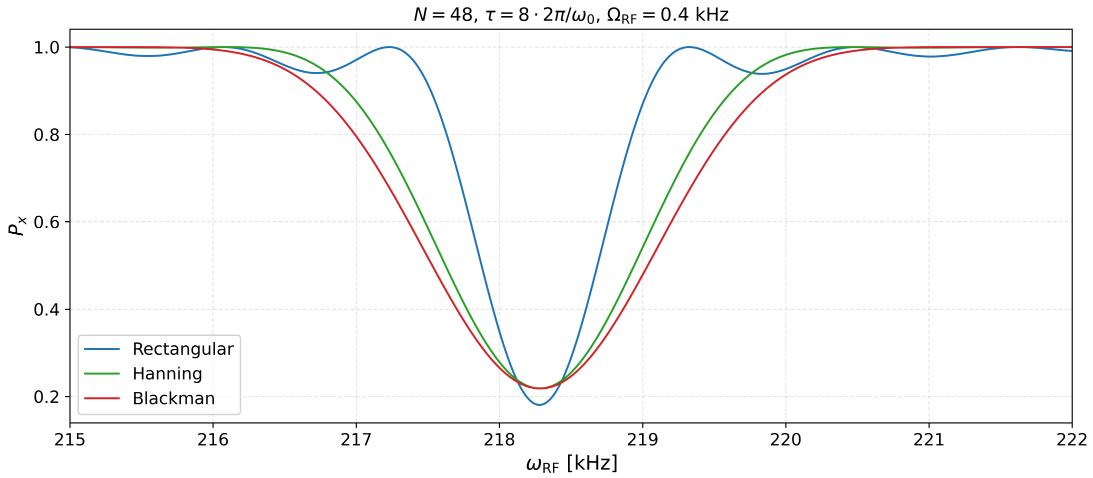
Intuition: the side-lobes are essentially the discrete Fourier transform of a rectangular window. Shaping the window suppresses its sidelobes, exactly the same trick used in classical signal processing.

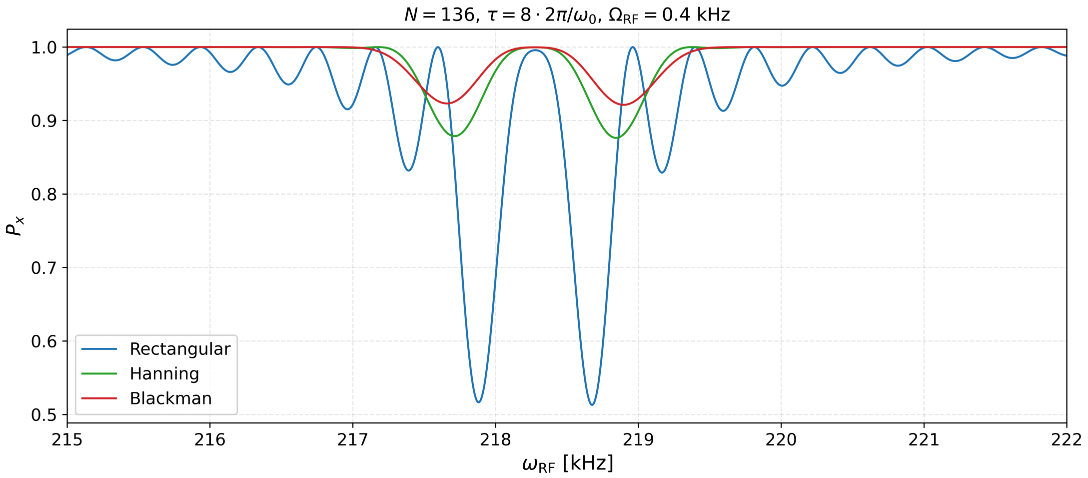
Suppression: Window Catalogue



The four windows we'll compare: rectangular (the baseline that produces the side-lobes), and three classical apodization windows from signal processing.

Suppression: Numerical Result





Apodized envelopes flatten the off-resonant region while preserving the on-resonant peak. The effect grows with N (right): rectangular develops a clean side-lobe pair, both apodized windows stay smooth.

Suppression: Window Comparison

The normalized spectral response factorizes into a window-independent `sinc` and a window-dependent kernel:

$$\left|\frac{F(\delta_1)}{F(0)}\right|^2 = \mathrm{sinc}^2(u) \cdot |G(u)|^2, \qquad u = \frac{\delta_1}{2\Omega \bar{f}}.$$

For example, while constant pulse gives : $\left|\frac{F}{F(0)}\right|^2_{\mathrm{rect}} = \mathrm{sinc}^2(u).$

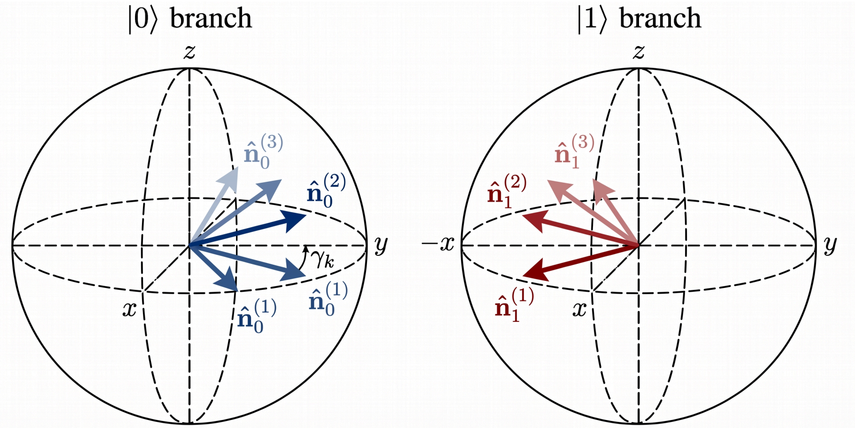
Blackman apodized pulse, $\left|\frac{F}{F(0)}\right|^2_{\mathrm{Black}} = \mathrm{sinc}^2(u) \cdot \left(\frac{50u^4 - 209u^2 + 84}{21(u^2-1)(u^2-4)}\right)^2$ where $u = \frac{\delta_1}{2\Omega \bar{f}}$.

Suppression: Build-up mismatch

The mismatch between consecutive axes is

$$\Delta\gamma_k \equiv \gamma_{k+1} - \gamma_k \approx -\frac{\delta_1}{\Omega}\Delta f_k$$

where $\Delta f_k = f(k+1) - f(k)$. So the non-commutativity enters at order $\delta_1 \cdot \Delta f_k$, which is small when either δ_1 is small or the window varies slowly.



Per-cell rotation axes: in-plane components antiparallel between branches; tilt γ_k drifts with k under apodization ($\Omega_k = \Omega\, f(k)$).

Outlook: Beyond Constant Ω_{RF}

The full-evolution formalism above assumes Ω_{RF} is time-independent (compatible with the RWA). With a Gaussian envelope

$$\Omega_{\mathrm{RF}}(t) = \Omega_0\, e^{-(t-t_k)^2/2\sigma^2},$$

the per-segment propagator $e^{-iH'_{0,1}\tau}$ is no longer exact. Direct Schrödinger integration takes tens of minutes per frequency point.

Approach: Magnus expansion, valid when Ω_{RF} varies slowly relative to ω_{RF}.

Outlook: Magnus Expansion

For $\partial_t U = -iH(t)U$, write $U = e^{\Omega(t)}$ with $\Omega(t) = \sum_n \Omega_n(t)$ where

$$\Omega_1 = \int_0^t A_1\, dt_1, \qquad \Omega_2 = \tfrac{1}{2}\iint [A_1, A_2]\, dt_1 dt_2, \ldots$$

Computed terms (with $f(t)$ the Gaussian envelope):

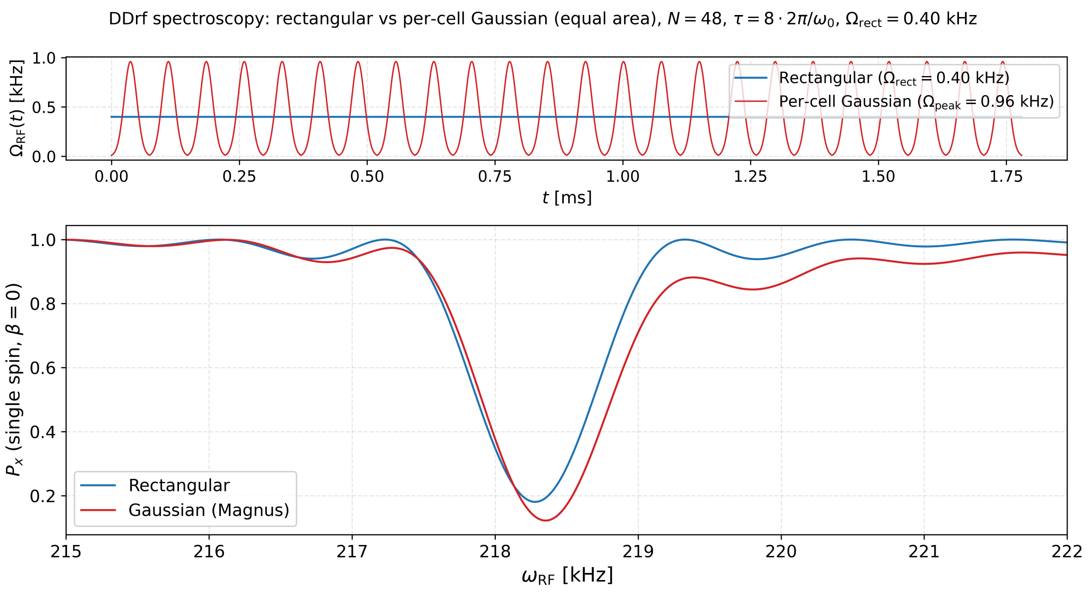
$$\Omega_1(T) = -i\Big(\delta_{(0,1)} T\, I_z + c_1(\cos\phi\, I_x + \sin\phi\, I_y)\Big)$$

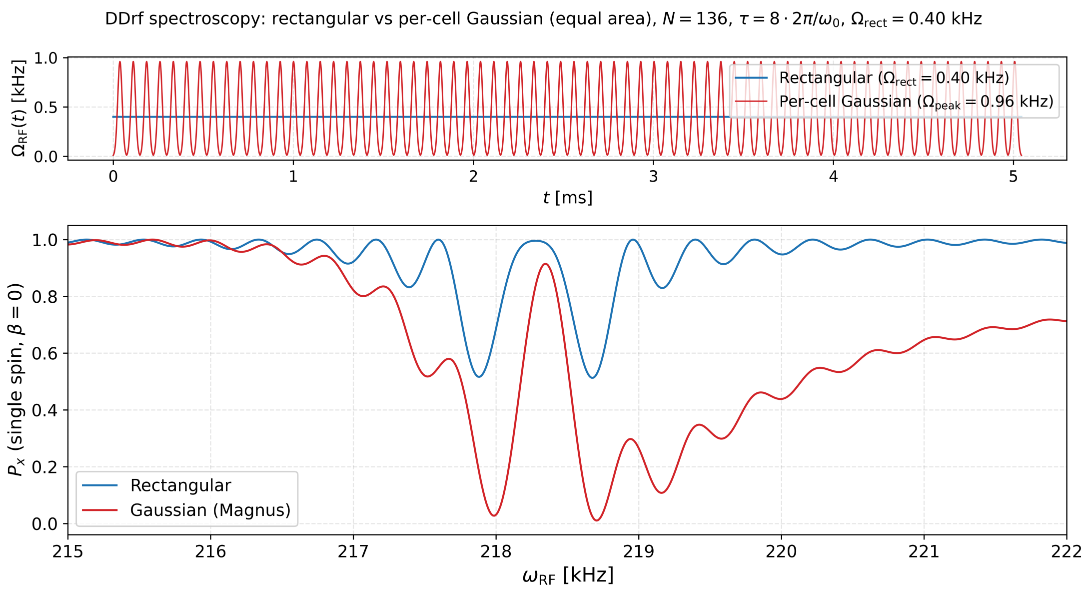
$$\Omega_3(T) = \frac{\delta_{(0,1)}^2}{24} K_1 \,(\cos\phi I_x + \sin\phi I_y) + \frac{\delta_{(0,1)}}{24} K_2\, I_z$$

$$\Omega_2 = \Omega_4 = 0$$

with $c_1 = \int_0^T f$ and $K_{1,2}$ triple integrals of f. Convergence is guaranteed when $\int_0^T \|A(s)\|_2\, ds < \pi$ — likely satisfied here; **simulation pending.**

Outlook: Gaussian Pulse Shaping — Spectroscopy





Thank You!

Questions & Discussion